Tharun Sridhar

AI Engineer | Bengaluru, India
+91 95916 47475 | tharunsridhar@gmail.com | LinkedIn | GitHub | Portfolio

Summary

AI engineer building full-stack LLM applications: multi-agent systems where specialized agents reason together, multimodal search that matches natural-language queries to images, and retrieval pipelines benchmarked across five different strategies. Comfortable owning a project from prompt design to a working FastAPI backend, tests included.

Technical Skills

Languages: Python, JavaScript, Java, SQL
AI / LLM Engineering: LLM Integration (Gemini, Groq, OpenAI), RAG, Agentic Tool-Calling, Model Context Protocol (MCP), Prompt Engineering, Structured Output Extraction, Vector Search, Hybrid Search (BM25+Vector), Reranking (RRF), Sentence-Transformers, LangChain Pipeline Design, LangGraph, Multi-Agent Frameworks (CrewAI, AutoGen, BeeAI)
Backend: FastAPI, Django, REST API Design, JWT Authentication, SQLAlchemy, Alembic, Testing (Pytest)
Databases: PostgreSQL, SQLite, ChromaDB (vector store), FAISS
DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS, Git, Linux

Education

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science — *May 2026*

Projects

Restaurant Recommendation System | *FastAPI, LangGraph, ChromaDB, CLIP* — GitHub

- Built a multimodal recommendation engine indexing restaurants and recipes by both text embeddings (Sentence-Transformers) and food image embeddings (CLIP), enabling cross-modal retrieval from natural-language dining requests.
- Designed a multi-agent LangGraph workflow with dedicated trend-analysis, cuisine/style-matching, and nutrition-fit agents that reason independently and synthesize into a single personalized recommendation.
- Exposed a full CRUD REST API (FastAPI, Pydantic) for restaurant and recipe management alongside the chat endpoint, with a pytest suite and a Procfile/runtime.txt for deployment.
- Implemented a Model Context Protocol server exposing restaurant lookup, vibe-based recommendation, and review tools plus a raw-text resource over stdio, with a client that discovers tools at runtime and a Gradio ReAct host wiring them into Groq function-calling.

Agent Framework Showdown | *FastAPI, CrewAI, AutoGen, BeeAI* — GitHub

- Built the same researcher-plus-writer task three separate ways across three multi-agent frameworks – a CrewAI sequential crew, an AutoGen `RoundRobinGroupChat`, and a BeeAI `ReActAgent` – behind one FastAPI service so their orchestration styles are directly comparable.
- Resolved a real cross-framework dependency conflict instead of avoiding it: CrewAI and BeeAI require incompatible pydantic versions, so each runs isolated in its own Python 3.11 venv as a subprocess communicating over stdout JSON, while AutoGen runs in-process on the main interpreter.
- Covered request validation and the venv subprocess wiring with a pytest suite that needs no API key or live model call, keeping fast automated tests separate from the live browser demo that exercises all three agents against Groq.

Retriever Showdown | *FastAPI, LangChain, Groq, BM25* — GitHub

- Built a full-stack app that runs the same query through five retrieval strategies at once – baseline dense vector search, MMR, LLM-driven multi-query rewriting, BM25 keyword ranking, and Reciprocal Rank Fusion – against an identical fixed corpus, making ranking differences directly comparable side by side.
- Reimplemented BM25 and RRF on the LangChain side (`BM25Retriever` plus a manual reciprocal-rank-fusion formula) instead of pulling in a second retrieval framework, keeping all five strategies on one stack behind one FastAPI backend.
- Built a frontend beyond a results viewer: each strategy panel carries a one-line explanation of its ranking logic plus an "Under the hood" view of the raw request/response JSON, surfacing the tradeoffs instead of just the output.

Open Source Contributions

django-guardian/django-guardian – fixed AUTO_PREFETCH permission bug for direct-permission models — #1030
drkrillo/good-first-issues – CI smoke test for the action itself — #172
Ayorinha/ayorai-vision-intelligence – restored corrupted CI workflow YAML — #26
Ayorinha/ayorai-vision-intelligence – MCP least-privilege regression tests — #23

Certifications

Model Context Protocol: Introduction & Advanced Topics — Anthropic
IBM RAG and Agentic AI Professional Certificate — IBM
Artificial Intelligence (with Google for Developers) — SmartBridge

Achievements

5⋆ HackerRank: Problem Solving, Java, Python, SQL — Profile
250+ LeetCode problems solved — Profile